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RESEARCH ARTICLE

Innovative Grouting Process Monitoring: Design and Implementation of a Measurement Device With Datalogger Function

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ABSTRACT Recording data from processes and activities has become indispensable for quality assurance and process verification, serving as a cornerstone in achieving research objectives. This requirement is particularly significant in the construction industry, where modern grouting techniques are used for waterproofing during tunnel wall renovations to safeguard structures from capillary water ingress, which is vital for their durability. Detailed documentation of grouting processes is often necessary, as these records offer valuable insights into substrate conditions and grouting quality. This paper focuses on designing and implementing a device that captures physical data during the grouting process and stores it in a defined format on a microSD card. The paper covers problem analysis, key input variables identification, and suitable hardware components selection. The proposed device is versatile, enabling real-time monitoring of the grouting process, and can be adapted to measure the consumption of other materials, such as one- or two-component paints. The device underwent rigorous testing under laboratory conditions to ensure its functionality and reliability, and thus significantly contributes to the control and responsiveness of the grouting process by providing real-time data on critical operational parameters. In addition to the controlled testing, the device prototype was subjected to real-world testing during a major grouting project at the Bratislava-Lamač railway tunnel in the Slovak republic. This large-scale construction site provided a valuable opportunity to assess the device's performance under operational conditions.

INDEX TERMS Grouting, sensor, microcontroller, datalogger, HMI.

I. INTRODUCTION

Recording data from processes and activities has become indispensable to ensuring quality and process verification in today's rapidly evolving construction industry. These records serve as indicators of performance and vital components for achieving research and operational objectives. A critical area where detailed documentation is necessary is the use of advanced technologies for construction projects, particularly

in techniques like modern grouting for waterproofing tunnel walls during renovation work [1]. Grouting plays a pivotal role in safeguarding structures against water ingress, which, if left unchecked, can severely compromise the longevity and integrity of a building. By capturing and maintaining thorough records of grouting processes, project managers and engineers can gain valuable insights into substrate conditions and the overall quality of the grouting operations, ensuring the long-term performance of the structure. Grouting is a specialized construction technique used to enhance the structural integrity of buildings, strengthen foundations,

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and provide protection against water intrusion. In many projects, especially those involving infrastructure improvements and large-scale construction, grouting becomes critical for foundation strengthening, allowing structures to achieve the necessary bearing capacity and stability to mitigate subsidence risks, landslides, or other ground-related issues [2]. In tunnel construction and repair, grouting creates a waterproofing barrier that prevents water from penetrating structural walls, thus shielding the building and its occupants from moisture-related damage [3]. This process is also essential in repairing structures like rock formations, where cracks and deformations are remediated to improve strength and resistance to unpredictable stresses and environmental changes. Moreover, grouting has significant applications in historical preservation. Many heritage buildings and aged infrastructure elements require restoration efforts to return to their original load-bearing capacity and waterproofing properties. In such cases, grouting offers an effective solution to restore stability while protecting the structures from further deterioration caused by water ingress. It not only prolongs the lifespan of these valuable landmarks but also ensures their continued use and preservation for future generations. This article aims to design and implement a device capable of capturing physical data from the grouting process and storing this information in a structured format on a microSD card. It will facilitate real-time monitoring of the grouting activity [4], providing critical data to ensure that the process is carried out effectively and complies with quality standards. The versatility of this device allows it to be adapted for recording other consumable substances, such as one- or two-component paints, making it a valuable tool across various applications in construction and engineering. The project is structured into three distinct parts. The first analyses the problem, where input variables and parameters to be recorded are identified. The second focuses on the hardware selection and evaluation of microcontroller boards and additional components required for the design. Finally, the software implementation explains the programming choices, focusing on the flow and logic needed to ensure seamless operation and data recording. For this research, the following research questions have been formulated:

A. How does the real-time monitoring capability of the device affect the accuracy and reliability of grouting process data compared to previous methods without such capabilities?

B. How does the integration of a data logger within the grouting process monitoring device influence data collection in the grouting process?

C. What potential improvements and future advancements are necessary to enhance the device's capabilities, particularly in terms of IoT integration?

II. RELATED WORK

Grouting is a critical renovation technique involving injection pumps to deliver specialized materials into a target location under controlled pressure [5], [6]. This technique consists

of injecting a grouting material through a pressure nozzle, which fills voids or cracks in a structure, reinforcing it and providing waterproofing capabilities. The pumps used in this process are available in various mechanical designs, including diaphragm, peristaltic, screw, or piston types. Each design offers specific advantages depending on the application, with drive types classified as electric, pneumatic, or electro-pneumatic. These configurations provide flexibility for different project scales and environments. During grouting, waterproofing material is injected into pre-drilled holes in the structure's walls through specialized pressure packers that activate in the cavity. The choice of grouting material varies, including single- and multi-component compounds and chemical compositions such as polyurethane, epoxy, and others [7]. Monitoring the material injection is crucial when using two-component polyurethane compounds, as the compound expands upon activation, potentially leading to localized damage if not carefully controlled.

Sensor monitoring plays a vital role in the grouting process, as it enables real-time control and assessment of critical parameters such as stability and strength characteristics. This monitoring is essential for extending the service life of structures and minimizing the risk of structural failures, accidents, or costly repairs. Regular analysis of the data collected during the grouting process allows for retrospective evaluations that inform future construction and technological procedures. Moreover, sensor systems provide technological and managerial benefits by addressing longstanding challenges in the construction industry. Historically, construction has been identified as an industry with low productivity, slow adoption of innovations, and poor economic efficiency [8]. However, integrating digital tools is a part of the principles of Industry 4.0 has gradually improved these issues. In particular, developing sensor systems within the Construction 4.0 framework has transformed process management by enabling real-time interconnection of construction activities [9].

Construction 4.0 encompasses various cutting-edge technologies, including the Internet of Things (IoT), artificial intelligence, big data, advanced analytics, and automation tools [10], [11]. These technologies enhance the efficiency and coordination of construction processes by allowing real-time data sharing and decision-making, thus increasing productivity and reducing waste. Furthermore, this development concept fosters innovations in construction materials and techniques that are more sustainable and efficient, addressing both environmental and economic concerns [12], [13]. From a quality control perspective, applying sensor systems ensures that projects meet the highest standards of quality, minimizing risks and reducing delays during construction. This streamlined process ultimately accelerates the commissioning of projects while offering insights that can drive future innovations in construction technologies. By identifying inefficiencies in previous methods and applying data-driven improvements, sensor systems enable the

evolution of more advanced technologies tailored to specific sectors within construction.

In terms of sustainability, introducing sensor systems aligns with environmental and economic goals. The construction industry faces growing pressure to reduce its environmental impact, particularly as increasing production volumes raise demand for raw materials and generate significant waste [14]. Effective monitoring of construction processes ensures the precise use of materials and energy, reducing waste and cost savings for investors. Moreover, digitalization within the construction sector enhances overall project management, enabling more accurate risk management, particularly for large infrastructure projects. Remote monitoring also improves safety by allowing workers to track the status of construction processes from a distance, reducing their exposure to dangerous environments and lowering the risk of accidents [15].

Despite the evident advantages of monitoring systems, the parameters that must be tracked during the grouting process are not always well-defined. In large-scale projects, these parameters are typically established by specialized construction engineers. However, achieving the set standards can be challenging due to unforeseen factors such as hidden cracks within building walls. Simple monitoring devices often fail to track critical grouting parameters such as flow rates and pressure levels. More advanced equipment incorporates sensors such as manometers or electronic systems, allowing operators to adjust grout dosage based on specific parameters [16]. Devices from leading manufacturers typically include both pressure sensors and flow meters for improved accuracy, and some models feature data loggers for comprehensive tracking of grouting operations [17]. Certain variants facilitate wireless transmission of measured data [18]. However, these advanced systems have limitations, such as their high cost, single-purpose design, and limited availability. While some practitioners suggest using programmable logic controllers (PLCs) to achieve similar functionality [19], this approach is often expensive and lacks the flexibility for future upgrades.

This article proposes the design of a grouting monitoring device built using widely available components to address these limitations. This device will be compatible with existing grouting equipment, enhancing its functionality. By analyzing the field data recorded by this device, it will be possible to create a database of materials and conditions for future grouting operations. The key physical parameters to be monitored include fluid flow rate, volume, and pressure during the injection process. The material temperature will also be measured to account for any volume changes due to temperature fluctuations. A real-time clock (RTC) will be included to ensure that time and date are accurately recorded for documentation purposes. This monitoring system will improve process control, ensuring that grouting operations meet the necessary standards and regulations.

The novelty of the proposed grouting monitoring system lies in its combination of affordability, flexibility, and

compatibility with existing grouting equipment. Unlike high-end, single-purpose commercial devices, which are often prohibitively expensive and difficult to upgrade, this system uses widely available components to provide advanced monitoring at a fraction of the cost. Its adaptability allows integration with various types of grouting machinery, enhancing their functionality without replacing or modifying existing setups. In addition to grouting, this system could be adapted to monitor other injection processes, such as applying paints or sealants, further extending its utility across different sectors in construction.

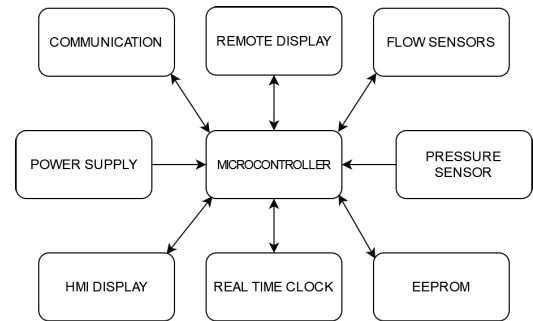


FIGURE 1. Device hardware architecture.

III. MATERIALS AND METHODS

This section of the proposal presents the complete hardware design of the device. To provide a more precise overview, we break down the overall circuit description into schematic blocks, explain the operating principles of each block, and implement the HMI (Human-Machine Interface) for the device. The hardware design proposal includes microcontroller board and eight modules constituting the complete device, as illustrated in Figure 1. The device is comprised of a central microcontroller board, which is responsible for reading and writing data from the connected peripherals, providing HMI functionality, and performing the necessary mathematical operations on the acquired data. Additional modules are tasked with relaying information regarding the measured physical parameters, such as flow sensors, pressure sensors, and the real-time clock. Another modules also handle the recording of the measured data to EEPROM and its visualization through the HMI and remote displays. The remaining modules, namely the power supply and communication blocks, serve to power the system and facilitate communication between components. This architecture ensures the effective operation of the device by integrating essential functionalities for data acquisition, processing, displaying, and system management. The device design was enhanced to enable remote display to show essential information on a smaller device that the operator can keep nearby during the grouting process. This device provides real-time data, allowing the operator to manage the process more precisely. The communication range between nodes can extend to several tens of meters. However, in real-world

conditions, electromagnetic induction may disrupt the signal from nearby sources of interference (such as power extension cords) or other electrical interferences. This problem is solved in the proposal.

A. MICROCONTROLLER BOARD

The entire device is controlled using the Arduino development platform. One of the key advantages of this platform is its extensive software libraries and widely recognized programming techniques [20], [21]. Arduino is available in various hardware configurations, which differ in the microcontroller used, size, communication interfaces, and specific hardware integrated directly onto the module board. The microcontroller board was also chosen because of the support of the ATSAM21G18A microcontroller [22]. Microchip's ATSAM21G18A microcontroller belongs to the SAM D21 family of low-power 32-bit Arm Cortex - M0 processors capable of operating at a clock frequency of 48MHz. Due to the size of the internal Flash memory of 256 kB and the static SRAM of 32 kB, it is suitable for applications that are more demanding on the size of the source code. It also has the advantage of a supply voltage range of 1.62 V - 3.63 V when using a battery power supply. Of course, I2C, UART, and SPI communication interfaces are also available. Despite its small size, the development environment offers a wide range of possibilities for development. An indisputable advantage is the presence of a micro-SD port for the memory card, which is connected via a separately dedicated serial peripheral interface (SPI) that we will use to write data to the memory card. This configuration is advantageous because it allows us to use the other ports for communication and peripherals. Naturally, Inter-Integrated Circuit (I2C), Universal Asynchronous Receiver-Transmitter (UART), and Serial Peripheral Interface (SPI) communication interfaces are also supported. Despite its compact size, the development environment provides many development opportunities. A significant advantage is the inclusion of a micro-SD port for the memory card, connected via a dedicated SPI interface. This interface is used to write data to the memory card, which is beneficial as it frees up the other ports for communication and peripheral connections.

B. POWER SUPPLY

Every electronic device requires a reliable power supply to operate properly, ensuring consistent voltage levels and protecting internal components from electrical anomalies such as overvoltage, undervoltage, or short circuits. These phenomena can damage sensitive components or cause unpredictable behavior in the system. Therefore, the power supply module must regulate and stabilize the input voltage, converting it into the required voltage levels for different sections of the device.

In the proposed system, the power supply module is designed to create three distinct voltage branches: +12 V, +5 V, and +3.3 V, all of which share a common ground,

as illustrated in Figure 2. Each voltage branch is critical for powering different parts of the device.

The +12 V and +5 V supply branches are generated using monolithic step-down converters, specifically the LM2575-5 and LM2575-12. These converters are efficient, fixed-output voltage regulators that step down the input voltage to provide stable +5 V and +12 V outputs, respectively. The converters are wired in accordance with the manufacturer's recommended configuration to ensure optimal performance, including proper filtering and component selection to minimize noise and ripple in the output voltage.

The 3.3 V power supply branch is regulated by the MCP1825-33 linear regulator, which derives its input directly from the +5 V source. Linear regulators like the MCP1825-33 are typically used when low-noise, stable voltage output is required, even though they are less efficient than switching regulators. In this design, the linear regulator ensures that the sensitive components requiring 3.3 V, such as microcontrollers, receive a clean, stable voltage without fluctuations that could affect their performance.

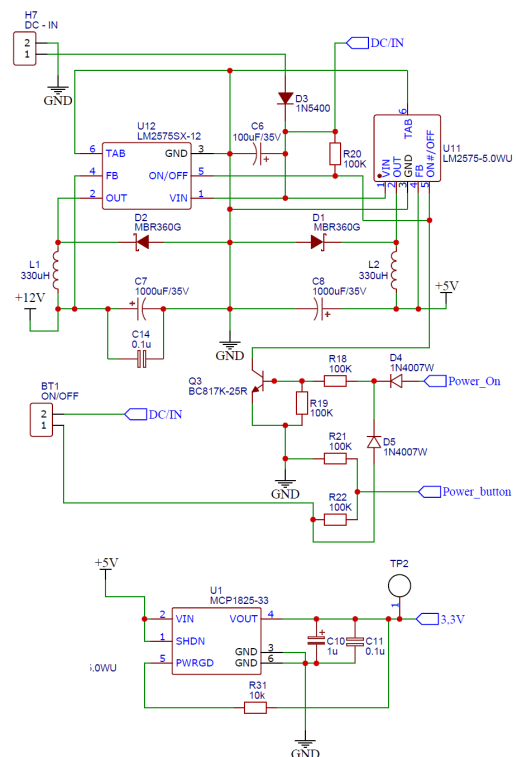


FIGURE 2. Power distribution circuitry.

C. COMMUNICATION

The device employs multiple communication interfaces to ensure data transfer between its various modules. In complex devices, where different components must work together efficiently, reliable communication channels are critical. These communication channels are crucial for complex devices integrating various functions and components into a single unit.

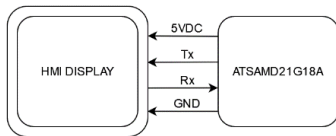


FIGURE 3. UART communication setup.

To manage communication with the Human-Machine Interface (HMI) display, as illustrated in Figure 3, the hardware-defined Serial Communication (SERCOM) module, specifically SERCOM1, was employed. The SERCOM1 interface is configured to operate as a standard UART (Universal Asynchronous Receiver/Transmitter) interface that facilitates the transfer of data between the device's microcontroller and the HMI. UART is chosen for its simplicity and effectiveness in serial communication, where data is transmitted bit by bit over a single line. This makes it ideal for interfacing with the HMI, as it supports the bidirectional transmission of data required for user input and system feedback.

The choice of a UART-based communication interface is particularly beneficial in this context because it offers reliable, low-overhead communication that does not require a high clock speed or complex handshaking protocols. UART enables the HMI to receive data from the system—such as real-time process parameters, alerts, and status updates—and also allows users to send commands or settings adjustments back to the system. This two-way communication is essential for real-time monitoring and control, making it a fundamental aspect of the device's interactive functionality.

In this proposal, the I2C interface facilitates the transfer of information between the microcontroller board, the Real-Time Clock (RTC) module, and the Electrically Erasable Programmable Read-Only Memory (EEPROM). The RTC module ensures accurate timekeeping, which is crucial for time-stamping data during the grouting process, while the EEPROM provides non-volatile storage for settings, calibration data, or system configurations that need to be retained even when the device is powered off. These modules will be described later.

D. SENSORS

The design incorporates several sensors crucial for accurate monitoring of the grouting process, specifically flow sensors and pressure sensors. For flow measurement, we have opted to use two separate flow meters, which allows for more precise and redundant flow measurements, improving the reliability and accuracy of the data collected. Each pulse flow meter type G-FLOW G0.4AL-HP measures the rate at which the grouting material is injected into the system, providing essential information on flow rates and helping to ensure the proper application of the grout.

The flow sensors in our design operate at a supply voltage of 12 VDC. To ensure that the signals from these sensors can be safely and accurately read by the microcontroller, we use optocouplers to isolate and manage the signals.

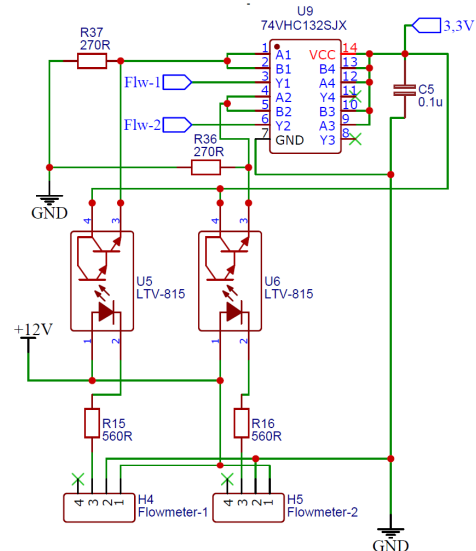


FIGURE 4. Flow sensor circuitry.

The optocouplers are driven by a current-limiting resistor to control the amount of current passing through the LED within the optocoupler. This isolation is beneficial in protecting the microcontroller and other sensitive components from high voltages or transient surges present in the sensor circuits.

As shown in Figure 4, a 560 Ω resistor is used in series with the optocoupler to regulate the current flowing into it. This resistor is chosen to limit the current to a nominal value of 20 mA, which is within the safe operating range of the optocoupler.

The device is monitoring the injection pressure by incorporating a Graco 243222 sensor that is installed in the fitting of the injection pump. This sensor measures the pressure of the grouting material as it is injected, providing data for monitoring and controlling the grouting process. However, the sensor's output signal, in its default state, is typically weak and may not be sufficient for accurate processing by the system's analog-to-digital converter (ADC).

To address this issue, an amplifier module is employed to boost the sensor's output signal. For this task, we have selected the AD620 instrumentation operational amplifier [23]. The AD620 is a precision, low-noise operational amplifier specifically designed for amplifying small signals, making it well-suited for use with pressure transducers. Its characteristics include high input impedance, low offset voltage, and low drift, which are crucial for accurately amplifying low-level signals from sensors. As shown in Figure 5, the AD620 is integrated into the design to enhance the pressure transducer's signal before it reaches the ADC.

E. MEMORY

The device was designed to handle input parameter adjustments, such as variations in flow sensors, by incorporating external memory to store critical data. This memory serves several key functions, including retaining calibration data

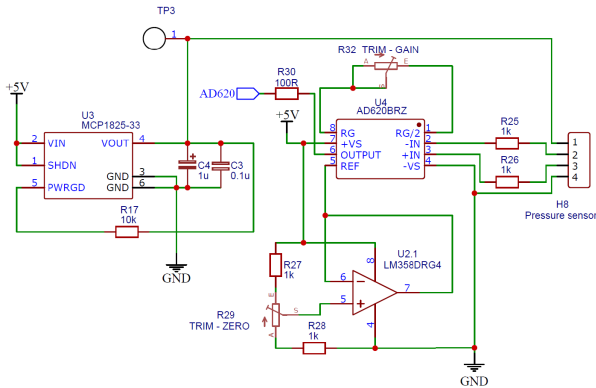


FIGURE 5. Pressure sensor circuitry.

from the flowmeter module, reference values from the pressure sensor's signal amplifier, and user-adjustable settings.

Initially, a microSD card was considered as a potential solution for recording and storing this data. However, the use of a microSD card presented a significant risk: during data transfers to a computer or other systems, there was a chance of inadvertently erasing or corrupting critical parameters. This potential for data loss or corruption was unacceptable for maintaining the integrity of the calibration and configuration data, which are essential for the accurate operation of the device.

To mitigate these risks, we opted for the Microchip Technology Inc. AT24C256 chip, a 256 Kbit I2C-compatible serial EEPROM, which provides a reliable and stable solution. This EEPROM chip offers 32 kilobytes of storage, organized into 32,000 8-bit words, making it well-suited for storing various types of data.

F. REAL-TIME CLOCK

The design includes a time and date recording system to ensure precise documentation of events and measurements. Accurate time-stamping is crucial for tracking the timing of specific events in the grouting process, such as the start and end of operations, as well as for recording any anomalies or adjustments made. This capability enables detailed logs, enhances data integrity, and facilitates effective analysis and reporting.

Several methods are available for implementing time measurement in embedded systems. One option is to utilize the Real-Time Clock (RTC) feature of the ATSAM21G18A microcontroller, which includes an integrated 32-bit counter with a 10-bit frequency divider. This built-in RTC can periodically generate time values, allowing the microcontroller to keep track of time and date with a certain degree of accuracy. However, while the ATSAM21G18A's RTC provides basic time-keeping functionality, it has limitations in terms of accuracy and stability. The internal RTC's timekeeping can experience deviations due to factors such as temperature variations or power supply fluctuations. These deviations may necessitate frequent time adjustments to

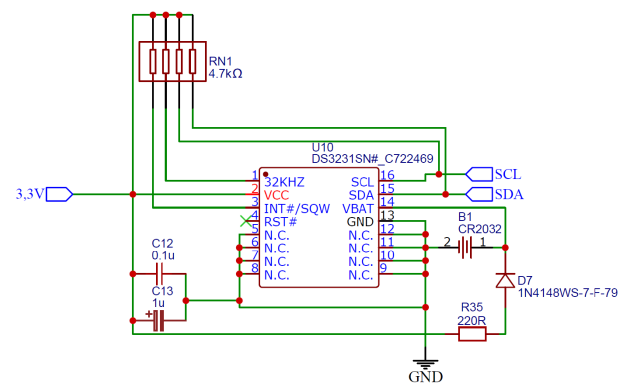


FIGURE 6. Real-time clock circuitry.

maintain accurate time records, which can be cumbersome and may introduce inconsistencies.

To address these limitations and ensure precise and reliable time measurement, we selected an external solution: the DS3231 Real-Time Clock module from Maxim Integrated, as depicted in Figure 6. The DS3231 is a high-precision RTC with built-in temperature compensation and battery backup, which communicates using I2C.

G. HMI DISPLAY

Incorporating a suitable display module significantly enhances the device by providing a clear, interactive interface for presenting all relevant operational details. The choice of display is crucial for user experience, particularly when it comes to visualizing and managing the injection process. The ability to clearly differentiate between various data frames and monitor the process effectively is paramount, which is why a color display unit was preferred over a monochrome one.

The NEXTION NX8048T070 HMI module [24] was selected to meet these requirements. This module features a 7-inch TFT LCD display with a resolution of 800 × 480 pixels, providing ample clarity and detail for the injection process. The color palette of 65,536 colors allows for vivid and easily distinguishable graphics, which enhances user interaction and understanding. Equipped with a resistive touch surface, the display enables users to interact directly with the information shown. The central unit of the NEXTION HMI module is powered by an STM32F family microcontroller running at 48 MHz. This microcontroller handles the display operations and ensures smooth performance of the graphical interface, efficiently managing complex user interface tasks. For storage, the module includes 16 MB of internal FLASH memory, which accommodates fonts, images, and other graphical assets, supporting detailed graphics and user interface elements. Additionally, it has 3584 bytes of RAM for managing runtime data and processing. Communication between the HMI module and the microcontroller board is facilitated via a Transistor-Transistor-Logic (TTL) serial link, which is reliable and straightforward, ensuring efficient data exchange

with the device's microcontroller. The module operates at 5 VDC and consumes 510 mA, a manageable requirement within the overall power budget of the device. An added feature of the module is its microSD port, which simplifies the transfer of compiled programs from the editor to the internal memory. This capability makes it easier to update or modify the user interface, ensuring that the latest graphical assets and functionalities are readily available.

H. REMOTE DISPLAY

For communication with the remote display panel, the RS485 communication bus was selected due to its excellent stability and sufficient transmission speed. This choice ensures reliable and consistent information transfer between devices, which is critical for maintaining accurate and up-to-date displays of operational data.

The ATMEGA32U4-AU microcontroller was chosen for the remote control unit. This microcontroller belongs to the family of 8-bit low-voltage Complementary Metal-Oxide-Semiconductor (CMOS) circuits based on the AVR architecture. The AVR architecture is enhanced by Reduced Instruction Set Computer (RISC) principles, which contribute to its efficient performance. The AVR core features 32 registers that are directly connected to the arithmetic logic unit, facilitating fast and efficient data processing. The microcontroller operates with a clock frequency of 16 MHz, providing adequate processing speed for handling communication tasks and controlling the remote display. The microcontroller is equipped with various memory resources: 1 kB of EEPROM for non-volatile data storage, 4 kB of SRAM for volatile data storage, and 32 kB of FLASH memory for program storage. The operating voltage range of the ATMEGA32U4-AU is between 2.7 VDC and 5.5 VDC, making it versatile for different power supply conditions. Additionally, the microcontroller is fully compatible with the Arduino bootloader environment, which facilitates ease of programming and integration within the Arduino ecosystem.



FIGURE 7. Information on the remote display.

For displaying the information transmitted from the host device, the Bolymin BL1602BMERNJU68J U2 alphanumeric display was selected. This display is equipped with an HD44780 controller, which is well-supported by libraries available in standard software development environments. The display presents crucial data from the system, including the injection point index and the amount of substance injected. This setup enables the operator to easily monitor and interpret the flowmeter data in real-time, as illustrated in Figure 7.

IV. RESULTS

The implementation of the device began with thorough individual testing of each peripheral. To validate the functionality of the various modules, a prototyping breadboard was utilized. During the prototyping phase, each component, including sensors, displays, and communication interfaces, was tested separately to ensure they met the required performance specifications. This preliminary testing was crucial for identifying and resolving any issues related to signal integrity, power requirements, and component interactions.

Once the individual modules were verified, the focus shifted to integrating these components into a cohesive system. A dedicated printed circuit board (PCB) was designed for the final prototype to streamline this integration. The design of the PCB, as illustrated in Figure 8, incorporated several surface-mounted parts (SMDs), which necessitated a more refined approach to assembly compared to the prototyping breadboard.

For the assembly of the PCB, a manual assembly process was employed. This involved placing each component onto the PCB by hand, which was essential for ensuring precision in the placement of the delicate SMDs. Reflow soldering technology was used to secure the components in place. The use of reflow soldering was particularly important due to the presence of numerous SMD components, which require precise and consistent soldering to ensure reliable operation.

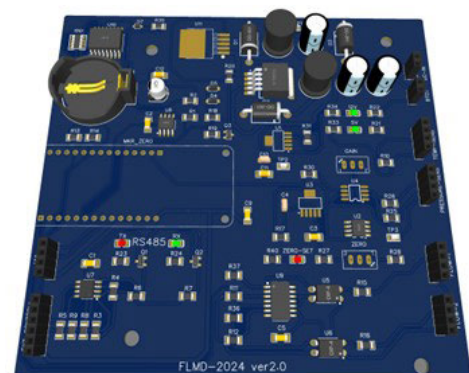


FIGURE 8. Prototype PCB visualization.

Once the device was fully assembled and tested, it was enclosed in a waterproof box to protect its components from environmental factors such as moisture, dust, and physical damage. This enclosure is crucial for ensuring the longevity and reliable operation of the device, particularly in challenging or outdoor environments where exposure to the elements is a concern. The printed circuit board (PCB), which houses the core electronic components and connections, is positioned at the bottom of the box. On the top of the box, the Human-Machine Interface (HMI) display is mounted, as illustrated in Figure 9.

Figure 10 displays the finalized device prototype enclosed within its waterproof enclosure. This design ensures that



FIGURE 9. Assembly of the device prototype.



FIGURE 10. Completed device in a box.

all internal components are protected from environmental factors such as moisture, dust, and physical impacts, which is crucial for maintaining the device’s reliability and durability in diverse conditions.

The control software is designed to manage communication with various peripherals, perform necessary

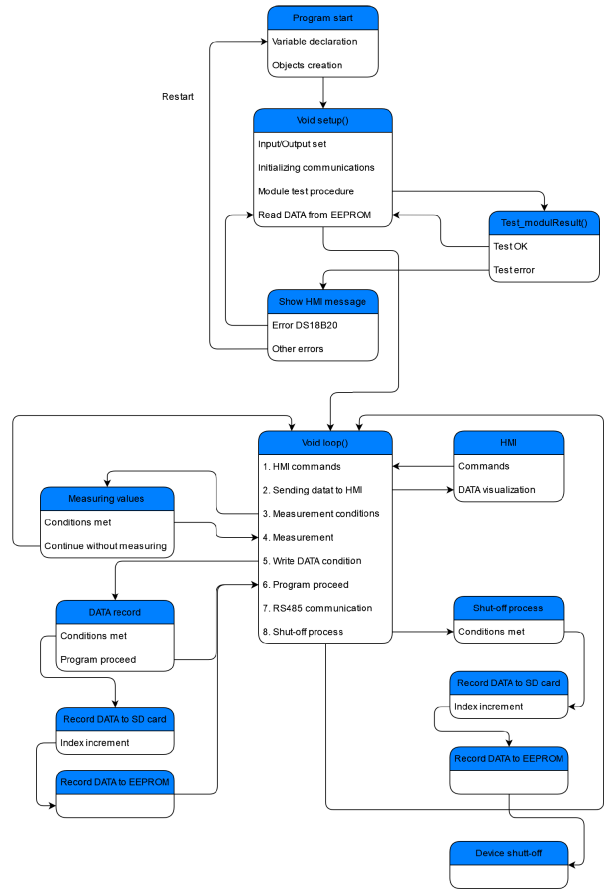


FIGURE 11. Control software diagram.

mathematical operations, and respond to user inputs. Additionally, it handles the critical tasks of data recording and information display. The software architecture is organized into distinct functional modules, each responsible for specific tasks. These functions are cyclically invoked based on predefined conditions. The runtime of the program can be divided into several phases, each corresponding to different operational states of the device. An overview of the program’s continuous loop, which is central to the device’s operation, is depicted in Figure 11. In the continuous loop, the program repeatedly cycles through its tasks, including monitoring sensor data, updating the display, processing user inputs, and recording data. This loop is designed to ensure that all functions are performed in a timely manner, maintaining the device’s responsiveness and accuracy.

The HMI display user interface is designed to operate autonomously by evaluating inputs according to predefined parameters. This design allows the device’s microcontroller unit to manage and adjust settings based on programmed algorithms. Users interact with the system through the HMI display, enabling real-time monitoring of data and adjustments to settings as needed.

The HMI interface screen, as depicted in Figure 12, is structured into multiple pages. Each page is organized to feature specific components that either display relevant

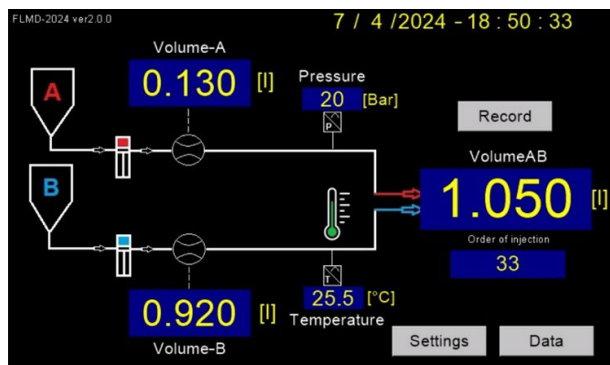


FIGURE 12. HMI display - main screen.



FIGURE 13. INTENSIFY test device connected with the proposed device.

information or facilitate user interactions using the virtual keyboard implemented for data entry, that includes exception handling in the programming code. This feature ensures that any parameters entered by the user adhere to the correct range and format.

V. DISCUSSION

The device underwent rigorous testing to ensure its functionality and reliability. Initially, we conducted tests under controlled conditions using specialized test equipment to validate the performance of the device. For the calibration of the pressure and flow sensors, we employed the INTENSIFY test device [25]. This device is specifically designed to simulate various pressure levels up to 6000 bars accurately using high-pressure hoses. By using the INTENSIFY test device [25], we could precisely calibrate the sensors and ensure they provided accurate readings under different pressure scenarios. For the purposes of this test, the upper pressure limit was set to 160 bar to align with the operational parameters of the proposed device.

The test procedure was divided into 16 incremental steps, each increasing by 10 bars. The pressure readings obtained from both the INTENSIFY test device and the proposed device are presented in Figure 13. This stepwise approach allowed for a detailed comparison between the reference values and

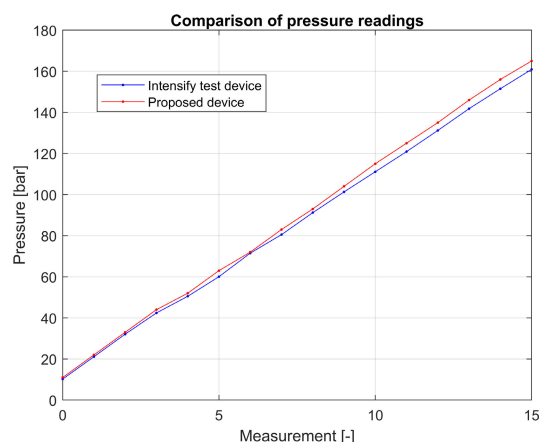


FIGURE 14. Comparison of pressure readings.

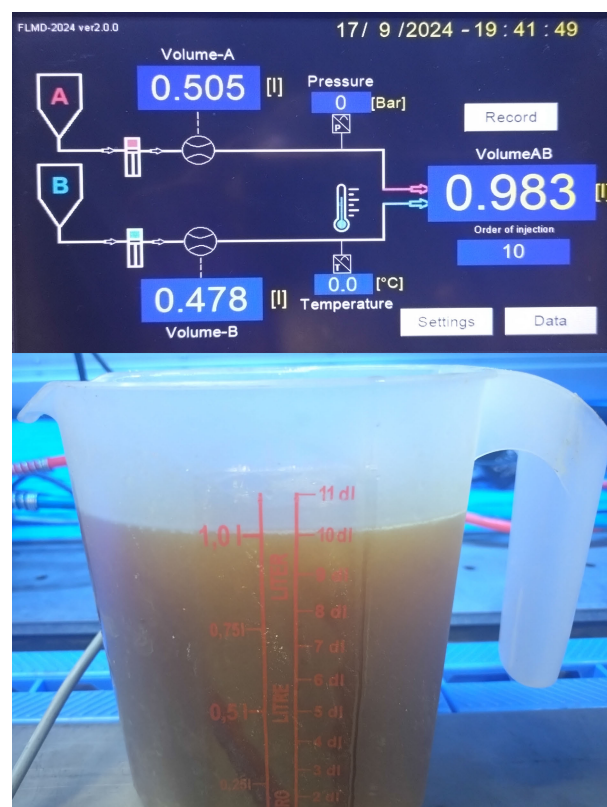


FIGURE 15. Comparison of pressure readings.

the measurements recorded by the device in Figure 14, ensuring the accuracy and consistency of the pressure sensor calibration across the tested range.

The pressure measurements within the natural gaps in the concrete do not rely on precise, absolute values, as these are difficult to estimate accurately. Instead, the data provides an indicative representation of the injection process and its progression. Consequently, the recorded values are presented as whole numbers, accompanied by the appropriate tolerance range, to reflect the inherent variability while offering meaningful insight into the behavior of the grouting process.

The subsequent test focused on flow measurements, as accurately determining the amount of injected material is critical for technological efficiency and economic considerations. It is essential to quantify the material applied at specific injection points and the total volume used throughout the process. Flow sensors with pulse output were utilized for this purpose, specifically for measuring viscous substances' flow under pressures of up to 400 bar. The flow sensors were tested in conjunction with the injection pump to verify the accuracy of the volume measurements. We employed calibrated measuring containers with 1 liter and 300 milliliters as reference points. The containers for components A and B were filled with liquid, and upon activating the pump, the containers were filled to their respective target volumes. The readings from the display were then compared to the volume indicated by the measuring containers. As shown in Figure 15 and Figure 16, the display values closely matched the reference values marked on the measuring containers, confirming the reliability of the flow measurement system.

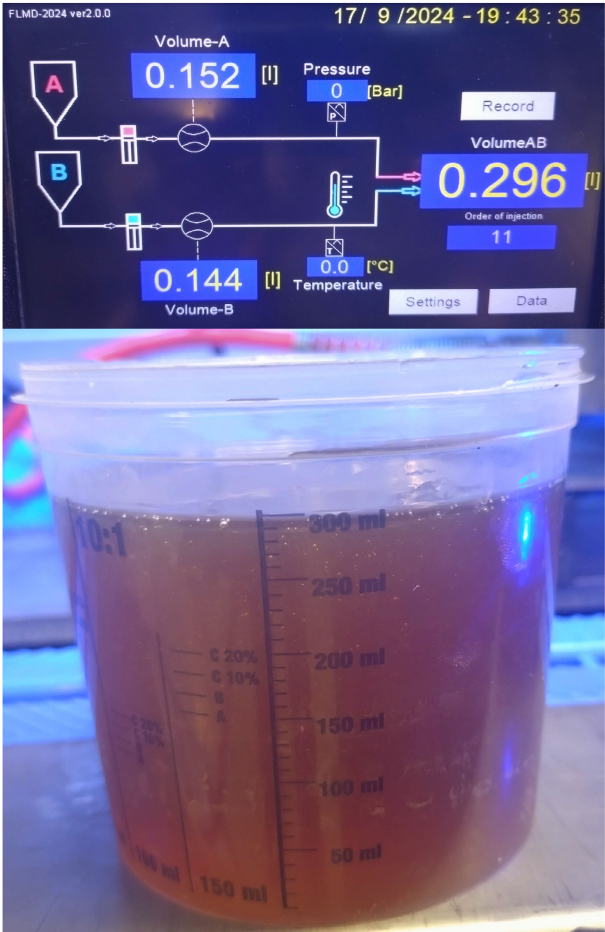


FIGURE 16. Comparison of pressure readings.

The measured values can also be monitored on an external display in Figure 17, which presents the cumulative data for components A and B.

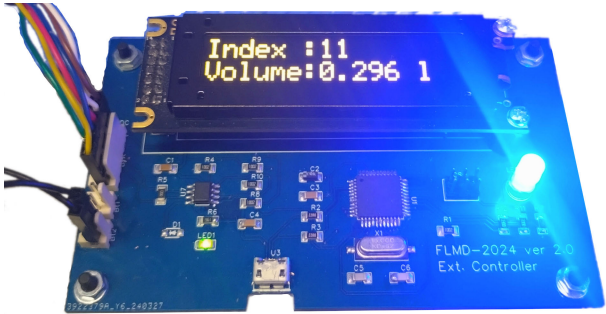


FIGURE 17. Comparison of pressure readings.

In addition to the controlled testing, the device prototype was subjected to real-world testing during a major grouting project at the Bratislava-Lamač railway tunnel in Slovak republic. This large-scale construction site provided a valuable opportunity to assess the device's performance under actual operational conditions. The grouting process involved executing approximately 40,000 grouting points, as illustrated in Figure 18 and Figure 19. These points were critical for creating an effective waterproofing barrier and ensuring the structural integrity of the tunnel.

Throughout this extensive field deployment, the device successfully recorded each grouting process, providing detailed data on the injection activities. The successful operation in such a demanding environment offered substantial insights into both the device's functionality and the grouting process itself. This real-world testing was instrumental in identifying areas for improvement and refining the device's design.

The feedback obtained from the field deployment was used to make significant enhancements to the device's current version. These refinements were aimed at improving the device's accuracy, reliability, and overall performance based on the insights gained from the actual use case in the tunnel. By integrating this field data into the design process, we ensured that the device was better suited to handle the complexities and challenges of grouting in real-world conditions.

The results of our research are summarized in Table 1. We could not compare the proposed solution with similar existing solutions due to their high cost and limited availability. Nevertheless, we believe that calibrating the device on a professional calibration machine provides strong evidence and confirms the validity of our design.

Future work on the device will focus on several key enhancements aimed at broadening its capabilities and aligning it with contemporary technological advancements.

One important direction for future development is the integration of additional sensors. Currently, the device is equipped to handle basic measurements, but adding sensors to monitor temperature and relative humidity, among other parameters, would significantly enhance its functionality. For instance, temperature sensors could track the temperature

TABLE 1. Research results.

Inputs	Observations	Outputs
Monitoring operational data during the grouting process	The real-time monitoring of key parameters such as pressure, flow rate, volume, and temperature enables precise adjustments to maintain process stability	Monitoring improves control and responsiveness of the grouting process
Recording operational data during the grouting process	The collected data allows for a retrospective analysis of the grouting process, as well as providing the basis for invoicing customers	The quantity of individual components from the used two-component grouting compound
Setting up calibration values using HMI display	The process allows for easy and direct interaction with the system, offering a clear interface for adjusting parameters such as pressure, flow rate, and temperature	By precisely adjusting parameters such as pressure, flow rate, and temperature, the grouting or operational process becomes more consistent and reliable
Device tests	Device tested under laboratory conditions	Consistent performance in different scenarios
Device deployment	Device deployed at the Bratislava-Lamač railway tunnel	Consistent performance in different scenarios

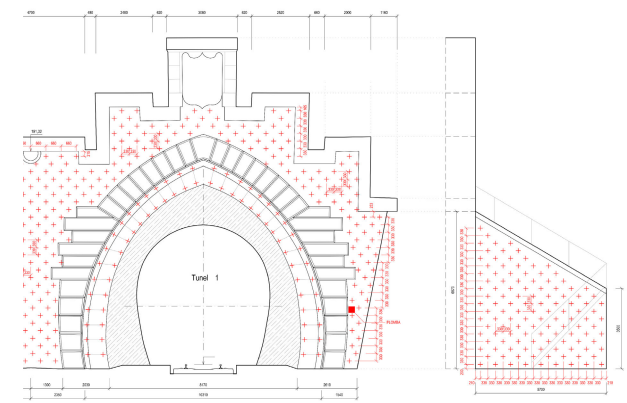


FIGURE 18. Grouting schematic of railway tunnel arch no.1.

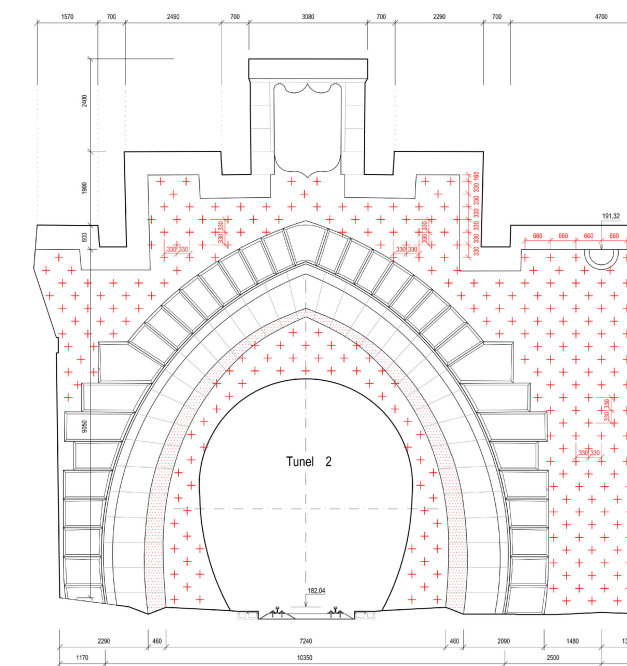


FIGURE 19. Grouting schematic of railway tunnel arch no.2.

of the grouting material, providing insights into its curing process and potentially affecting its overall performance. Similarly, relative humidity sensors could help monitor

environmental conditions that impact the grouting operation. By expanding the range of measurable factors, the device would offer a more comprehensive analysis of the grouting process, leading to better process control and quality assurance.

Another critical area for development is upgrading the device to an Internet of Things (IoT) capable system. This upgrade would involve enabling the device to connect to a designated server or cloud-based platform, allowing for real-time data storage and analysis. IoT integration would provide numerous advantages, such as enhanced monitoring capabilities, where operators and project managers could access live data remotely. This real-time access would facilitate better oversight of the grouting process and enable timely interventions when necessary.

Storing data on a centralized server would enable advanced data analytics and reporting. Historical data could be analyzed to identify trends, optimize processes, and make more informed decisions. Overall, these enhancements will make the device more versatile, responsive, and integrated with modern digital infrastructure, thereby increasing its value and applicability across various construction and engineering projects.

The proposed device significantly advances the functionality of conventional pressure grouting pumps used in construction by integrating advanced inspection and recording capabilities. Traditional grouting pumps, while effective for their primary function, typically lack the sophisticated monitoring and data recording features that are increasingly valuable in modern construction practices. While there are existing solutions that offer similar functionalities, these are often high-cost systems tailored for specific applications, limiting their accessibility and flexibility.

Our proposed device addresses these limitations by offering a more adaptable and cost-effective solution. Unlike specialized, high-priced systems, this device provides a versatile configuration that can be easily adjusted to meet varying requirements in different grouting applications. It supports comprehensive data recording, capturing detailed information throughout the grouting process, which can then be exported for further analysis or documentation. This capability ensures that users can maintain thorough records,

assess performance, and make informed decisions based on accurate and real-time data.

The proposed device significantly contributes to the control and responsiveness of the grouting process by providing real-time data on critical operational parameters. It allows operators to make immediate adjustments, ensuring the process remains within optimal conditions throughout its duration. For example, monitoring the quantity of individual components in the two-component grouting compound ensures that the right proportions of materials are used, avoiding inconsistencies or errors that could affect the quality of the grout or the overall project. Another contribution is in the device calibration. By precisely adjusting parameters such as pressure, flow rate, and temperature, operators can ensure the grout is injected at the correct force, delivered at the right speed, and cured under the most suitable conditions. These adjustments lead to more predictable and stable outcomes, reducing the chances of defects or inefficiencies. Accurate control over these parameters minimizes material waste and improves resource utilization, making the grouting process more efficient and cost-effective. As a result, the overall reliability of the process is greatly enhanced, and the risk of process failures or undesirable results is significantly reduced.

The effectiveness of the device has been demonstrated through its practical application during construction grouting projects. It has been successfully deployed in real-world scenarios, such as the grouting operations at the Bratislava-Lamač railway tunnel, where it performed reliably and provided valuable data. The positive results from these applications highlight the device's practicality and its potential for routine use in various construction settings.

With only minor adjustments needed for different operational contexts, the device is poised to become a valuable tool in the construction industry. Its ability to enhance existing grouting systems with advanced monitoring and recording features represents a significant improvement over conventional methods. By offering a flexible, user-friendly, and cost-effective solution, the device stands out as a practical alternative to more specialized and expensive equipment.

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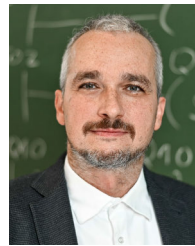
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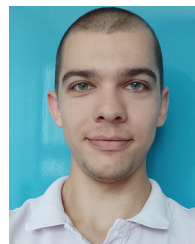


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